

(a) Finite-amplitude recovery

What requested disturbance amplitude remains recoverable?



Path: $s_i(t) \geq s_{\min}$

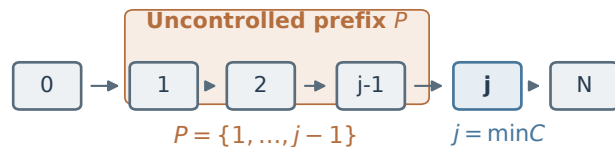
Terminal: $v_i \rightarrow \bar{v}$, $s_i \rightarrow \bar{s}_i$, $a_i \rightarrow 0$

Loss: $\bar{j} \leq \ell$

Deadline: T_r

System-level limit α .

(b) Directed prefix obstruction



Required safety-bound violation

Downstream actuation cannot change
this prefix trajectory

Fixed leader path, delayed histories,
parameters, active set and directed topology

**Prefix safety violation →
conditional policy-independent obstruction**

(c) Complementary evidence

**NONLINEAR
SAME-MODEL WITNESS**

Fixed leader path/history

Heterogeneous nonlinear
dynamics

Logged positive-gap
prefix crossing



**Policy-independent
obstruction (fixed run)**

Distinct nonlinear and sampled-linear evidence

**SAMPLED-LINEAR
RESERVE**

$$Q = Q_0 + \alpha L_{\pi} d$$

$$m_r = b_r - c_r^T Q_0$$

$$\gamma_r = h_D(L_{\pi}^T c_r)$$

$$\alpha_{\pi, \text{aff}} = \min \{ \alpha_{\max}, \min_{r: \gamma_r > 0} m_r / \gamma_r \}$$

Limiting row:
vehicle $\times$ time $\times$ constraint
Conditional upper bound

(d) Spatial implication

Four active CAVs in each nonlinear $N = 40$ layout

Front empty P

Even empty P

Middle **93/102**
witnesses

Rear **105/107**
witnesses

Equal tail transfer can coexist with different
local constraint reserves (Supplementary Fig. S2).